

Refinement of Reliability Grid Codes in the Provision of Ancillary Services

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NOMENCLATURE

Functions

$c_{t,i}(\cdot)$ Cost function of reserve provider i at hour t
 $F(\cdot)$ Cumulative distribution function of available reserve

Sets

$\mathcal{I}$ Reserve providers, $i \in \mathcal{I} = \{1, \dots, |\mathcal{I}|\}$
 $\mathcal{T}$ Hours, $t \in \mathcal{T} = \{1, \dots, |\mathcal{T}| = 24\}$
 Ω Scenarios, $\omega \in \Omega = \{1, \dots, |\Omega|\}$
 Ω' Out-of-sample scenarios, $\omega' \in \Omega' = \{1, \dots, |\Omega'|\}$

Parameters

$\alpha_{t,i}, \beta_{t,i}$ Intercept and slope coefficients of the cost function for reserve provider i at hour t [€/MW]
 $1 - \varepsilon^{\text{sys}}$ Threshold for system-level reserve reliability [-]
 $\pi_{t,\omega}$ Probability of scenario ω at hour t
 ρ^{sys} Penalty for reserve capacity shortfall compromising system security at hour t [€/MW]
 ρ^{viol} Penalty for reserve capacity shortfall at hour t [€/MW]
 D_t Minimum reserve to be procured at hour t [MW]
 $R_{t,i}, r_{t,i,\omega}$ Random variable representing the reserve capacity availability of provider i at hour t , and its realization under scenario ω [MW]

Primal variables

$1 - \varepsilon_t$ Threshold for reserve reliability at hour t [-]
 $\hat{b}_{t,i}$ Reserve capacity bid offered by provider i at hour t [MW]
 $b_{t,i}$ Cleared reserve capacity of provider i at hour t [MW]
 d_t Cleared reserve demand at hour t [MW]
 $s_{t,\omega}^{\text{sys}}$ System-level reserve shortfall compromising system reliability under scenario ω at hour t [MW]
 $s_{t,\omega}^{\text{viol}}$ System-level reserve shortfall under scenario ω at hour t [MW]
 $s_{t,i,\omega}$ Provider i 's reserve shortfall under scenario ω at hour t [MW]
 $y_{t,\omega}$ Binary variable indicating system failure, where $y_{t,\omega} = 1$ if a reserve capacity shortfall occurs under scenario ω , and $y_{t,\omega} = 0$ otherwise

Dual variables

$\underline{\mu}_{t,i}, \bar{\mu}_{t,i}$ Dual variables associated with the bounds of the reserve bid by provider i at hour t
 $\underline{\nu}_{t,i}, \bar{\nu}_{t,i}$ Dual variables associated with the bounds on the cleared reserve capacity of provider i at hour t
 λ_t Reserve market-clearing price at hour t [€/MW]

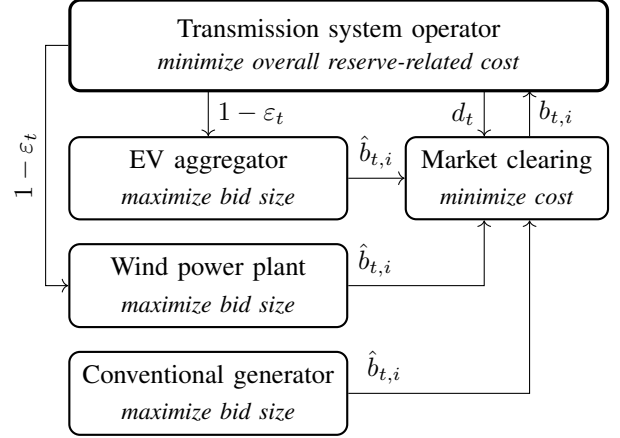


Fig. 1: Information exchange in the bi-level optimization formulation. In the upper level, the TSO sets the reliability threshold $1 - \varepsilon_t$, optimized hourly in the dynamic case and reduced to a single value $1 - \varepsilon$ in the static case, to minimize total reserve-related costs, including reserve procurement and expected shortfall penalties. In the lower level, EV aggregators and wind power plants receive the threshold and solve chance-constrained problems to maximize their reserve bids $\hat{b}_{t,i}$ based on probabilistic forecasts, while conventional generators submit bids constrained only by available capacity. The reserve market-clearing problem then determines the cleared reserve capacities $b_{t,i}$ given reserve demand d_t and all submitted bids, which are anticipated by the TSO when determining the optimal reliability threshold.

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With the set of decision variables $\xi \in \{\varepsilon_t, d_t, s_{t,i,\omega}, s_{t,\omega}^{\text{sys}}, s_{t,\omega}^{\text{viol}}, y_{t,\omega}, z_{t,i}, \hat{b}_{t,i}, \bar{\mu}_{t,i}, \underline{\mu}_{t,i}, b_{t,i}, \bar{\nu}_{t,i}, \underline{\nu}_{t,i}, \lambda_t\}$, $\forall t, i, \omega$,

the single-level optimization problem is (1a)-(1t):

$$\begin{aligned}
\min_{\xi} \quad & \overbrace{\sum_{t,i} c_{t,i}(1-\varepsilon_t)b_{t,i}}^{\text{reserve provision cost}} \\
& + \overbrace{\sum_{t,\omega} \pi_{t,\omega} (\rho^{\text{viol}} s_{t,\omega}^{\text{viol}} + \rho^{\text{sys}} s_{t,\omega}^{\text{sys}})}^{\text{reserve shortfall cost}}, \quad (1a) \\
s_{t,i,\omega} \geq 0, \quad & \forall t, i, \omega, \quad (1b) \\
s_{t,i,\omega} \geq b_{t,i} - r_{t,i,\omega}, \quad & \forall t, i, \omega, \quad (1c) \\
\sum_i s_{t,i,\omega} = s_{t,\omega}^{\text{viol}}, \quad & \forall t, \omega, \quad (1d) \\
d_t - s_{t,\omega}^{\text{viol}} + M y_{t,\omega} \geq D_t, \quad & \forall t, \omega, \quad (1e) \\
|\Omega| - \sum_{\omega} y_{t,\omega} \geq |\Omega|(1 - \varepsilon^{\text{sys}}), \quad & \forall t, \quad (1f) \\
y_{t,\omega} \in \{0, 1\}, \quad & \forall t, \omega, \quad (1g) \\
s_{t,\omega}^{\text{sys}} \geq 0, \quad & \forall t, \omega, \quad (1h) \\
s_{t,\omega}^{\text{sys}} \geq D_t - d_t + s_{t,\omega}^{\text{viol}}, \quad & \forall t, \omega, \quad (1i) \\
0.8 \leq (1 - \varepsilon_t) \leq 1, \quad & \forall t, \quad (1j) \\
\hat{b}_{t,i} \leq z_{t,i}, \quad & \forall t, i, \quad (1k) \\
-\hat{b}_{t,i} \leq 0, \quad & \forall t, i, \quad (1l) \\
-1 + \bar{\mu}_{t,i} - \underline{\mu}_{t,i} = 0, \quad & \forall t, i, \quad (1m) \\
-\bar{\mu}_{t,i} z_{t,i} \geq -\hat{b}_{t,i}, \quad & \forall t, i, \quad (1n) \\
b_{t,i} \leq \hat{b}_{t,i}, \quad \forall i, \quad & (1o) \\
-b_{t,i} \leq 0, \quad & \forall t, i, \quad (1p) \\
d_t - \sum_i b_{t,i} = 0, \quad & \forall t, \quad (1q) \\
\alpha_{t,i} + (1 - \varepsilon_t)\beta_{t,i} - \underline{\nu}_{t,i} + \bar{\nu}_{t,i} - \lambda_t = 0, \quad & \forall t, i, \quad (1r) \\
\sum_i -\bar{\nu}_{t,i} \hat{b}_{t,i} + \lambda_t d_t \\
\geq \sum_i (\alpha_{t,i} + (1 - \varepsilon_t)\beta_{t,i}) b_{t,i}, \quad & \forall t, \quad (1s) \\
\bar{\mu}_{t,i}, \underline{\mu}_{t,i}, \bar{\nu}_{t,i}, \underline{\nu}_{t,i} \geq 0, \quad \lambda_t \text{ free}, \quad & \forall t, i. \quad (1t)
\end{aligned}$$

where $F^{-1}(\frac{\varepsilon_t}{0.2}) = z_{t,i}$ for notational simplicity.

The Weibull CDF is defined as:

$$F(\hat{b}_{t,i}) = \begin{cases} 1 - \exp(-\kappa(\hat{b}_{t,i})^\gamma), & \hat{b}_{t,i} \geq 0, \\ 0, & \hat{b}_{t,i} < 0. \end{cases} \quad (2)$$

APPENDIX

A. Reformulation of chance constraint calculations

Denoting the available reserve at the 20th percentile by $r_{t,i}^{0.2}$, i.e., $\mathbb{P}(R_{t,i} \leq r_{t,i}^{0.2}) = 0.2$, where, as before, $R_{t,i}$ is a continuous random variable following a distribution depending on the forecast of the individual reserve provider i at hour t , we express the conditional probability of a realization $r_{t,i,\omega}$

of the random variable $R_{t,i}$ being under the reserve providers placed bid $\hat{b}_{t,i}$ using Bayes' theorem:

$$\begin{aligned} & \mathbb{P}(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r_{t,i}^{0.2}) \\ &= \frac{\mathbb{P}(R_{t,i} \leq \hat{b}_{t,i}) \mathbb{P}(R_{t,i} \leq r_{t,i}^{0.2} \mid R_{t,i} \leq \hat{b}_{t,i})}{\mathbb{P}(R_{t,i} \leq r_{t,i}^{0.2})} \end{aligned} \quad (3a)$$

$$\Rightarrow \mathbb{P}(R_{t,i} \leq \hat{b}_{t,i}) = 0.2 \mathbb{P}(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r_{t,i}^{0.2}), \quad (3b)$$

where it is implicit that $\mathbb{P}(R_{t,i} \leq r_{t,i}^{0.2} \mid R_{t,i} \leq \hat{b}_{t,i}) = 1$, and $\mathbb{P}(R_{t,i} \leq r_{t,i}^{0.2}) = 0.2$ as previously stated. We then have:

$$1 - \mathbb{P}(R_{t,i} \leq \hat{b}_{t,i}) \geq 1 - \varepsilon_t, \quad (3c)$$

which by (3b) and some basic arithmetic is the same as:

$$0.2 \mathbb{P}(R_{t,i} \leq \hat{b}_{t,i} \mid R_{t,i} \leq r_{t,i}^{0.2}) \leq \varepsilon_t. \quad (3d)$$

B. Convexity of the CDF

For $\hat{b}_{t,i} \geq 0$ we investigate the convexity conditions of the Weibull CDF (2) by the second derivative:

$$F'(\hat{b}_{t,i}) = f(\hat{b}_{t,i}) = \kappa \gamma (\hat{b}_{t,i})^{\gamma-1} \exp(-\kappa(\hat{b}_{t,i})^\gamma) \quad (4a)$$

$$\begin{aligned} F''(\hat{b}_{t,i}) = f'(\hat{b}_{t,i}) = & \kappa \gamma (\gamma - 1) (\hat{b}_{t,i})^{\gamma-2} \exp(-\kappa(\hat{b}_{t,i})^\gamma) \\ & - \kappa^2 \gamma^2 (\hat{b}_{t,i})^{2(\gamma-1)} \exp(-\kappa(\hat{b}_{t,i})^\gamma). \end{aligned} \quad (4b)$$

$F(\hat{b}_{t,i})$ is strictly convex for all $\hat{b}_{t,i} \geq 0$ if $F''(\hat{b}_{t,i}) > 0$ or strictly concave if $F''(\hat{b}_{t,i}) < 0$. Canceling common terms in (4b), namely $\exp(-\kappa(\hat{b}_{t,i})^\gamma)$, $\kappa \gamma$, and $(\hat{b}_{t,i})^{\gamma-2}$, which are always non-negative, we find the convexity of $F(\hat{b}_{t,i})$ by the sign of the remaining terms:

$$\text{sign}(\gamma - 1 - \kappa \gamma (\hat{b}_{t,i})^\gamma). \quad (4c)$$

$F(\hat{b}_{t,i})$ is either convex or concave depending on the sign of $F''(\hat{b}_{t,i})$ which solely depends on γ . When $\gamma \leq 1$, $F(\hat{b}_{t,i})$ is globally concave. When $\gamma > 1$, $F(\hat{b}_{t,i})$ is convex, then concave, with the inflection point:

$$\hat{b}_{t,i}^* = \left(\frac{\gamma - 1}{\kappa \gamma} \right)^{1/\gamma}, \quad (4d)$$

i.e., $F(\hat{b}_{t,i})$ is convex on $(0, \hat{b}_{t,i}^*)$ and concave on $(\hat{b}_{t,i}^*, \infty)$.

C. Convexity of the inverse CDF

Let $F : (a, b) \xrightarrow{\text{onto}} (c, d) \subset \mathbb{R}$ be a convex function and let $F^{-1} : (c, d) \rightarrow \mathbb{R}$ be its inverse. Then, if F is increasing then F^{-1} is increasing and concave [1]. This intuition also applies in the other direction, from concave to convex. Due to the nature of CDFs, we know that F is an increasing function, and since the function is convex on $(0, \hat{b}_{t,i}^*)$ and concave on $(\hat{b}_{t,i}^*, \infty)$, we know that its inverse is concave on $(0, \hat{b}_{t,i}^*)$ and convex on $(\hat{b}_{t,i}^*, \infty)$.

D. McCormick envelopes

McCormick envelopes are a type of convex relaxations of bilinear optimization problems with continuous variables. Given a bilinear expression $u_{t,i} = (1 - \varepsilon_t)b_{t,i}$ where each variable is continuous and has defined lower and upper bounds from (1j), (1o) and (1p), the new objective function with the relaxations are defined as:

$$\begin{aligned} \min_{\xi} \quad & \sum_{t,i} \alpha_{t,i} b_{t,i} + \beta_{t,i} (1 - u_{t,i}) \\ & + \sum_{t,\omega} \pi_{t,\omega} (\rho^{\text{viol}} s_{t,\omega}^{\text{viol}} + \rho^{\text{sys}} s_{t,\omega}^{\text{sys}}) \end{aligned} \quad (5a)$$

with added constraints:

$$u_{t,i} \geq 0.8b_{t,i}, \quad \forall t, i, \quad (5b)$$

$$u_{t,i} \geq b_{t,i} + (1 - \varepsilon)\hat{b}_{t,i} - \hat{b}_{t,i}, \quad \forall t, i, \quad (5c)$$

$$u_{t,i} \leq b_{t,i}, \quad \forall t, i, \quad (5d)$$

$$u_{t,i} \leq (1 - \varepsilon)\hat{b}_{t,i} + 0.8b_{t,i} - 0.8\hat{b}_{t,i}, \quad \forall t, i. \quad (5e)$$

This method reduces the complexity of the objective function, at the cost of introducing more bilinear terms in the constraints. The method is applied to all bilinear terms in (1). For dual variables, bounds are found analytically in preparation for the relaxation.

E. Linearizations of F

As $F^{-1}(\varepsilon_t/0.2)$ is nonlinear, we linearize it for numerical simplicity. In the feasible and relevant domain of $1 - \varepsilon_t \in [0.8, 1]$, we make n evenly spaced cuts and linearize between them, which Gurobi handles on its own. Hence, there is no need to implement SOS2 [2] or other methods. The definition of $F^{-1}(\varepsilon_t/0.2)$ is such that it may take negative values which is a direct violation of (11). The linearization enables us to remove the equality condition in $F^{-1}(\frac{\varepsilon_t}{0.2}) = z_{t,i}$ which is causing the violation.

REFERENCES

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- [2] S. A. Gabriel *et al.*, *Complementarity Modeling in Energy Markets*. Springer New York, 2013.